

Chapter 2. PERMUTATIONS

2.2 Cycles



Today we will prove the

CYCLE DECOMPOSITION THEOREM

THEOREM 2.17. CYCLE DECOMPOSITION THEOREM.

Every permutation $\sigma \neq \varepsilon$ (in S_n) is a product of (one or more) disjoint cycles of length at least 2.

Proof. PART 1: Existence

Take $\sigma \in S_n$ such that $\sigma \neq \varepsilon$. We proceed by induction on $n \geq 2$.

Basis step. If $n=2$, then we are done because σ is a transposition.

Inductive step Assume that the result is true for permutations in S_{n-1} and we prove it for $\sigma \in S_n$.

If $\sigma(n) = n$, then we can view σ as a permutation in S_{n-1} , and the result follows by the inductive hypothesis.

Assume now that $\sigma(n) \neq n$, and let $k = \sigma^{-1}(n)$. Then

$$\underline{\underline{\sigma(k) = \sigma(\sigma^{-1}(n)) = \underline{\underline{n}},}}$$

So $k \neq n$ because $\sigma(n) \neq n$. Thus, we can consider the transposition $\gamma = (k \ n)$ and the permutation $\tau = \sigma\gamma$.

From $\gamma^2 = E$, multiplying on $\tau = \sigma\gamma$ on the left by γ we obtain that

$$\boxed{\sigma = \tau\gamma} \leftarrow$$

Moreover, $\underline{\tau(n)} = \sigma(\gamma(n)) = \sigma(k) = \underline{\underline{n}}$.

From here we can view τ as an element of S_{n-1} , and by the inductive hypothesis, τ can be written as a product of disjoint cycles.

We consider two cases:

Case 1 . $\tau(K) = K$

Notice that τ fixes both K and n , which are the only elements of $\mathbb{Z}_n$ moved by γ ($\gamma = (K \ n)$). That is to say that τ and γ are disjoint permutations. Then

$$\sigma = \tau \gamma = \underline{\tau(K \ n)} \text{ is a product}$$

product of
disjoint
cycles

of disjoint cycles, finishing the proof of this case.

Case 2. $\tau(K) \neq K$

Since τ is a product of disjoint cycles, we have that K is moved by exactly one of the cycle factors of τ .

Thus: $\tau = \mu(\underbrace{K \ K_1 \ \dots \ K_r})$,

where μ is a product of disjoint cycles fixing $K, K_1, \dots, K_r$, and n because we proved that $\tau(n) = n$.

Therefore:

$$\sigma = \tau\sigma = \mu(K \ K_1 \ \dots \ K_r)(K \ n)$$

$\stackrel{(*)}{=} \mu(K \ n \ K_1 \ \dots \ K_r)$ is a product of disjoint cycles.

$$\textcircled{*} \quad \underbrace{((K) \xrightarrow{\quad} K_1 \dots K_r)}_{\alpha} \underbrace{(\xrightarrow{\quad} n)}_{\beta} \stackrel{?}{=} \underbrace{(K \xrightarrow{\quad} n \xrightarrow{\quad} K_1 \dots K_r)}_{\beta}$$

$$\alpha: K \mapsto n \mapsto n$$

$$\alpha: n \mapsto K \mapsto K_1$$

$$\alpha: K_i \mapsto K_i \mapsto K_{i+1}$$

$$i = 1, \dots, r-1$$

$$\alpha: K_r \mapsto K_r \mapsto K$$

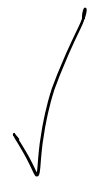
$$\beta: K \mapsto n$$

$$\beta: n \mapsto K_1$$

$$\beta: K_i \mapsto K_{i+1}$$

$$i = 1, \dots, r-1$$

$$\beta: K_r \mapsto K$$



PART 2, Uniqueness of the factorisation

$$\begin{aligned}\text{Suppose that } \sigma &= \tau_a \dots \tau_2 \tau_1 \\ &= \delta_b \dots \delta_2 \delta_1\end{aligned}$$

where the τ_i and δ_j are disjoint cycles.

GOAL : To prove that
$$\begin{cases} a = b \\ \text{and} \\ \tau_i = \delta_i \text{ for} \\ \text{all } i \text{ (after} \\ \text{possible reordering)} \end{cases}$$

We proceed by induction on $t = \max\{a, b\}$.

If $t = 1$, there is nothing to prove because $\sigma = \tau_1 = \delta_1$.

Suppose now that the result is true for permutations admitting two factorisations into disjoint cycles with less than t factors.

$$\sigma = \gamma_a \dots \gamma_2 \gamma_1 = \delta_b \dots \delta_2 \delta_1$$

$$t = \max \{a, b\}.$$

Let σ move m . Since the γ_i 's and δ_j 's are disjoint cycles, then m appears in exactly one of the γ_i 's and in one of the δ_j 's. Reordering (if necessary) we can assume that m occurs in γ_1 and in δ_1 .

We can write

$$\gamma_1 = (m \ k_2 \ \dots \ k_r)$$

$$\delta_1 = (m \ l_2 \ \dots \ l_s)$$

Assume (for example) that $r \leq s$.

Then

$$k_2 = \gamma_1(m) = \sigma(m) = \delta_1(m) = l_2$$

$$\begin{aligned} k_3 &= \gamma_1(k_2) = \sigma(k_2) = \sigma(l_2) = \\ &= \delta_1(l_2) = l_3 \end{aligned}$$

$\vdots$

$$\underline{k_r = l_r}$$

If $r < s$, then

$$\begin{aligned} \underline{m} &= \gamma_1(k_r) = \sigma(k_r) = \sigma(l_r) = \\ &= \delta_1(l_r) = \underline{l_{r+1}} \end{aligned}$$

But then we have that

$$\delta_1 = (\underline{m} \ l_2 \ \dots \ l_r \ \underline{m} \ \dots \ l_s),$$

which is impossible because δ_1 is a cycle. Therefore $r = s$ and so

$$\underline{\gamma_1 = \delta_1}.$$

$$\text{Write } \mu = \gamma_1 = \delta_1.$$

$$\begin{aligned}\sigma &= \tau_a \dots \tau_2 \tau_1 = \delta_b \dots \delta_2 \delta_1 \\ &= \delta_b \dots \delta_2 \tau_1\end{aligned}$$

or $\boxed{\sigma = \tau_a \dots \tau_2 \mu = \delta_b \dots \delta_2 \mu}$

From here we get that

$$\sigma \mu^{-1} = \tau_a \dots \tau_2 = \delta_b \dots \delta_2$$

is a product of $a-1$ (and $b-1$) disjoint cycles. By the inductive hypothesis we obtain that

$$a-1 = b-1 \Rightarrow \boxed{a=b}$$

and

$$\tau_i = \delta_i \text{ for } i=1, \dots, a-1$$

